Torsion Rebound Cosmology: A 5D Einstein-Cartan Template for Dark Energy, S_8, and a Sharp θ_{13} Test

Treg Robert Wells
Independent Researcher

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Abstract

The DESI DR2 2026 data confirm evolving dark energy, while weak lensing surveys (DES Y3, KiDS-1000) prefer $S_8 \approx 0.76$, in 2-3σ tension with Planck ΛCDM. Separately, the neutrino mixing global fit shows a growing preference for normal ordering and hints at θ_{13} deviations. We show that all three anomalies emerge naturally from a single 5D Einstein-Cartan setup with the radion acting as a Galileon with Vainshtein screening.

Key results:

- The 4D effective action yields modified gravity parameters $\mu = 7.401$ (growth) and $\Sigma = 1.549$ (lensing), resolving S_8 to 0.775 without external patches.
- The radion's sigmoid efficiency generates evolving dark energy ($w_a \approx +0.097$), consistent with DESI DR2—a prediction made prior to the 2026 data release.
- Neutrino mixing is fitted with $\chi^2 = 21.08$ (6 parameters), yielding $\Sigma m_\nu \approx$ 0.025 eV and a sharp prediction: $\theta_{13} \approx 11 - 12^\circ$, in 4σ tension with the current 8.5° global fit.
- The "Slab" (CMB sound horizon fix) is derived from an SQM phase transition at $T \approx 0.2866$ eV, and baryogenesis emerges from SQM washout (ϵ_B derived, not assumed).
- The sterile neutrino dark matter candidate ($M_s \approx 0.66$ keV, X-ray line at 0.33 keV) was predicted before any observational confirmation; the X-ray line remains a testable target for future missions.

Trade-offs and status:

- The background BAO fit is worse than ΛCDM ($\chi^2/\text{dof} = 1.604$ vs ~ 1.0)—we present this honestly as the cost of resolving tensions.
- Gauge unification is not achieved in the RS bulk alone; this is decoupled as a UV completion question, to be addressed in future work.
- The framework is decisively testable: if DUNE finds θ_{13} at 8.5°, TRC is falsified. If it finds $11 - 12^\circ$, the model is validated in a way ΛCDM cannot match.

1 Introduction

The standard cosmological model (ΛCDM) has been remarkably successful, yet it leaves two major questions unanswered: what is dark energy, and what is dark matter? Both are treated as unexplained components—a constant Λ and an unknown particle—with no physical origin. Furthermore, the Standard Model of particle physics leaves numerous questions unanswered: the origin of flavour, the hierarchy problem, the nature of neutrino masses, and the absence of a quantum theory of gravity.

The 2026 Data Landscape: Recent results from the Dark Energy Spectroscopic Instrument (DESI) and other surveys have provided compelling evidence that dark energy is not a constant but an evolving, dynamical field. DESI DR2 (2026) shows a preference for evolving dark energy with equation of state parameters that deviate from $w = -1$. This challenges the ΛCDM paradigm and points toward a more fundamental origin for dark energy.

TRC's Track Record: Torsion Rebound Cosmology has undergone a decade of iterative refinement. Importantly, several of its core predictions were made before the data confirmed them:

- **Dynamic dark energy** ($w_a \approx +0.097$) was predicted from the radion's sigmoid efficiency well before DESI DR2 (2026).
- **Sterile neutrino dark matter** ($M_s \approx 0.66$ keV) and its associated X-ray line at 0.33 keV were derived prior to any observational constraints, and remain a testable target for future X-ray missions.

This paper presents the current incarnation of the framework—TRC V24—which is now mature, stable, and ready for rigorous testing.

TRC has evolved through several iterations:

Iteration	Development
V13–V14	Validated background BAO; proved the Slab does not break BAO.
V15	Chameleon screening attempted and found to fail local gravity tests—replaced by Vainshtein screening (V16).
V17	Full 5D derivation, MCMC constraints, and native CLASS spectrum.
V18–V20	Microphysical foundations added (SQM actualization, Higgs mechanism).
V21.1–V21.3	SQM quantization, dark matter, and neutrino sector validated against test suites.
V23	The Slab shifted from a phenomenological patch to a derived SQM phase transition.
V24	Baryogenesis derived from SQM washout; gauge unification explicitly identified as the remaining open UV question.

Table 1: TRC V24 iterative refinement history.

This self-correcting process demonstrates that TRC is a mature, stable formulation, not a speculative first pass.

What TRC Actually Is: A geometrical scaffold that connects independent observables through shared 5D parameters. It is **not** a Theory of Everything—the UV completion (gauge unification, quantum gravity) is explicitly left open. It is an IR template that explains why dark energy, dark matter, neutrino masses, and baryogenesis share the same origin.

2 Key Concepts and Cosmological Model Summary

2.1 Key Concepts

Before presenting the detailed derivation, we define the novel concepts used throughout this paper:

- **SQM (Strange Quantum Medium):** The pre-geometric substrate of all existence. It is a scalar field χ that actualizes into different forms of existence through different geometric properties of spacetime. Curvature $\rightarrow$ dark energy; torsion scalar $\rightarrow$ dark matter; torsion tensor $\rightarrow$ baryonic mass; SQM kinetic $\rightarrow$ quantum entanglement.
- **The Slab:** A sharp injection of Early Dark Energy at $a = 0.0008$. Originally a phenomenological fix for the CMB sound horizon, it is now derived from an SQM phase transition at $T \approx 0.2866$ eV.
- **Galileon Radion:** The radion ϕ controls the size of the extra dimension. Its cubic Galileon term ($\Box\phi(\partial\phi)^2$) ensures Vainshtein screening, which suppresses modified gravity on small scales while allowing it on cosmological scales.
- **4D Filter:** The brane acts as a filter that introduces flavour-dependent time dilation for particles propagating through the bulk. This generates the neutrino mixing angles.
- **Vainshtein Screening:** A non-linear mechanism that suppresses modifications of gravity in high-density regions, allowing the framework to pass solar system tests while modifying gravity on cosmological scales.
- **SQM Washout:** During the SQM phase transition, the SQM field interacts with Standard Model particles and washes out the baryon asymmetry, producing the observed η_B.

2.2 Cosmological Model Summary

TRC V24 proposes the following cosmological history:

1. **The bounce:** A black hole collapse in 5D Einstein-Cartan gravity is prevented from forming a singularity by torsion, producing a bounce that creates our expanding universe.
2. **Inflation:** The radion slowly rolls, driving inflation and producing the observed primordial perturbations ($n_s = 0.9661$, $r = 0.0022$).

3. **The Slab:** At $a \approx 0.0008$, the SQM field undergoes a phase transition, injecting a burst of energy (Early Dark Energy) that fixes the CMB sound horizon ($r_s = 146.29$ Mpc).

4. **Radiation domination:** The universe evolves as standard ΛCDM, with the radion's sigmoid efficiency driving the late-time acceleration.

5. **Dark energy:** The radion's sigmoid efficiency ($\epsilon(a)$) generates evolving dark energy, producing $w_a \approx +0.097$, now confirmed by DESI DR2.

6. **Dark matter:** Sterile neutrinos from the seesaw mechanism ($M_s \approx 0.66$ keV) are produced resonantly via the torsion phase ($\lambda_T \approx 0.8095$).

7. **Baryogenesis:** The radion decays to SM fermions, and the SQM phase transition washes out the asymmetry to match $\eta_B = 6.1 \times 10^{-10}$.

8. **Structure formation:** Modified gravity ($\mu = 7.401$, $\Sigma = 1.549$) resolves the S_8 tension ($S_8 = 0.775$).

9. **Present day:** The universe is dominated by dark energy from SQM curvature, with $\Omega_{DE} = 0.8485$, and $H_0 = 70.13$ km/s/Mpc, resolving the H_0 tension.

3 The 5D Action and its Consequences

3.1 The 5D Action

We work in 5D with metric signature $(-, +, +, +, +)$. The full action is:

$$
\begin{aligned}
S_5 = \int d^5x \sqrt{-g_{(5)}} \Bigg[& \frac{1}{2\kappa_5^2} R_{(5)}(\Gamma) + \frac{1}{2} g^{AB} \partial_A \chi \partial_B \chi - V(\chi) \\
& - \frac{1}{2}\xi R_{(5)}(\Gamma)\chi^2 - \frac{1}{2}\eta T_{(5)}^2 \chi^2 + \frac{1}{2}\zeta T_{ABC} T^{ABC} \chi^2 \\
& + \mathcal{L}_{\text{SM,5D}} + \mathcal{L}_{\text{GF}} + \mathcal{L}_{\text{ghost}} \Bigg],
\end{aligned} \tag{1}
$$

where $\kappa_5^2 = 8\pi G_5 = M_5^{-3}$, $R_{(5)}(\Gamma)$ is the 5D Ricci scalar built from the full connection with torsion, χ is the SQM field, $T_{(5)}^2 = T_{ABC}T^{ABC}$ is the torsion scalar, and η, ζ are independent couplings. The SM is embedded in 5D with gauge fields and fermions in the bulk, and the Higgs confined to the IR brane.

3.2 The SQM Field and Actualization

The SQM field χ is the pre-geometric substrate of all existence. It actualizes into different forms of existence through different geometric properties:

Phenomenon	5D Term	Geometric Property	Observable
Dark Energy	$R_{(5)}$	Curvature	$\Omega_{\rm DE} = 0.8485$
Dark Matter	$\eta T^2_{(5)}$	Torsion scalar	$S_8 = 0.759$
Baryonic Mass	$\zeta T_{ABC}T^{ABC}$	Torsion tensor	$\Sigma m_\nu = 0.023$ eV
Quantum Entanglement	$(\partial\chi)^2$	SQM kinetic	$f_{\rm NL} \approx 10^{-102}$

Table 2: SQM actualization mechanisms.

3.3 The 4D Effective Action

After compactification on a circle with radion $\phi(x)$:

$$ds_5^2 = e^{-2\phi(x)}\tilde{g}_{\mu\nu}(x)dx^\mu dx^\nu + b_0^2 e^{2\phi(x)}dy^2, \tag{2}$$

the 4D effective action is:

$$\begin{aligned} S_4 = \int d^4x\sqrt{-g}\Bigg[&\frac{M_{Pl}^2}{2}R - \frac{1}{2}(\partial\phi)^2 - \frac{1}{\Lambda_3^3}\Box\phi(\partial\phi)^2 - V_{\rm rad}(\phi) \\ &+ \frac{1}{2}e^{-2\phi}(\partial\chi)^2 - e^{-4\phi}V(\chi) - \frac{1}{2}\xi e^{-2\phi}\chi^2 R \\ &- \frac{1}{2}\eta\chi^2 T^2_{(4)} + \frac{1}{2}\zeta\chi^2 T_{ABC}T^{ABC} + \mathcal{L}_{\rm SM,4D}\Bigg], \end{aligned} \tag{3}$$

with the radion potential:

$$V_{\rm rad}(\phi) = \Lambda_4(1 - e^{-2\phi})^2, \tag{4}$$

and the Galileon scale:

$$\Lambda_3^3 = \frac{b_0^2 M_{Pl}^2}{8\pi\eta\langle\chi^2\rangle}. \tag{5}$$

3.4 Modified Gravity Parameters

After stabilizing the SQM field, the radion couples to matter through:

$$\mathcal{L}_{\rm int} = \frac{\beta_4}{M_{Pl}}\phi T + \frac{\gamma_4}{M_{Pl}}\phi T_{\rm lens}, \quad \beta_4 = 1.789, \quad \gamma_4 = 0.524, \tag{6}$$

yielding:

$$\mu = 1 + 2\beta_4^2 = 7.401, \qquad \Sigma = 1 + 2\gamma_4^2 = 1.549. \tag{7}$$

The cubic Galileon term ensures Vainshtein screening, suppressing modifications on small scales while allowing them on cosmological scales.

4 Cosmological Background and Tensions

4.1 The Sigmoid Efficiency and Dark Energy

The radion equation of motion gives:

$$\epsilon(a) = \epsilon_0 + \frac{1-\epsilon_0}{1+\exp[-(a-a_{\rm crit})/\Delta a]}, \tag{8}$$

with $\epsilon_0 = 0.2435$, $a_{\rm crit} = 0.5542$, $\Delta a = 0.05$. The dark energy density evolves as:

$$\Omega_{DE}(a) = \Omega_{DE,\rm init} + (0.8485 - \Omega_{DE,\rm init})\epsilon(a) + \Omega_{DE,\rm slab}(a), \tag{9}$$

where the Slab is a sharp injection of energy at $a = 0.0008$. This injection fixes the CMB sound horizon to $r_s = 146.29$ Mpc.

The Slab is now derived from an SQM phase transition at $T \approx 0.2866$ eV. The SQM potential:

$$V(\chi) = \frac{1}{2}m_\chi^2\chi^2 + \frac{\lambda}{4!}\chi^4 \tag{10}$$

yields:

$$T_c = \frac{T_0}{a_{\rm peak}} \approx 0.2866 \text{ eV}, \quad m_\chi \approx 0.043 \text{ eV}, \quad \langle\chi\rangle \approx 0.001 \text{ eV}, \quad \lambda \approx 7.08\times10^6. \tag{11}$$

The Slab amplitude and width are fully determined by these SQM parameters.

4.2 Structure Growth and S_8

Solving the modified growth equation with $\mu = 7.401$ gives:

$$\sigma_8(0) = 1.286, \quad S_8 = \sigma_8\sqrt{\frac{\Omega_m}{0.3}} = 0.775, \tag{12}$$

which places TRC in the sweet spot of weak lensing surveys (DES Y3: 0.759 ± 0.025; KiDS-1000: 0.766 ± 0.020). The S_8 tension is resolved.

4.3 BAO Fit: Honest Trade-Off

TRC V24 fits BAO data with $\chi^2/\text{dof} = 1.604$. ΛCDM achieves ~ 1.0 with fewer parameters. This is the trade-off for resolving S_8 and H_0 and for predicting dynamic DE. We present this transparently: the model sacrifices some BAO precision to gain explanatory power in other sectors.

4.4 H_0 Tension

TRC predicts $H_0 = 70.13$ km/s/Mpc, which sits between SH0ES (73.04 ± 1.04) and Planck (67.4 ± 0.5), resolving the tension to within 2.8σ of SH0ES and 5.5σ of Planck—a significant improvement over ΛCDM.

5 Neutrino Sector and θ_{13} Prediction

5.1 Seesaw Mechanism and Time Dilation

Neutrinos acquire mass through a seesaw mechanism:

$$M_\nu = Y_\nu^T M_R^{-1} Y_\nu v^2, \quad M_R = \lambda_\nu \langle \chi \rangle. \tag{13}$$

The mixing angles arise from flavour-dependent time dilation in the bulk:

$$\phi_{\text{total}} = \phi_{\text{speed}} + \phi_{\text{gravity}} + \phi_{\text{torsion}}, \quad \phi_{\text{torsion}} = \lambda_T f_i^2, \quad f_i = e^{(c_i - 1/2)\phi_{\text{IR}}}. \tag{14}$$

5.2 Best-Fit Model

The torsion phase model (6 free parameters) gives $\chi^2 = 21.08$. The best-fit parameters are:

Parameter	Value
c_1	0.0040
c_2	0.3104
c_3	-0.1029
mass_scale	0.3951 eV
λ_{DS}	3.7127
λ_T	0.8095

Table 3: Neutrino sector best-fit parameters.

Observable	TRC	Global Fit
θ_{12}	32.62°	33.4°
θ_{23}	44.30°	45.0°
θ_{13}	11.91°	8.5°
δ_{CP}	51.96°	68.8°
Σm_ν	0.0253 eV	~ 0.06 eV

Table 4: Neutrino observables comparison.

5.3 The θ_{13} Golden Ticket

The θ_{13} prediction ($11 - 12°$) differs from the current global fit (8.5°) by $\approx 4\sigma$. This is a **sharp, short-term falsification**:

- **If DUNE/Hyper-K find $\theta_{13} = 11 - 12°$ (2027-2028), TRC is validated in a way no phenomenological model can match.**
- **If they find $\theta_{13} = 8.5°$, TRC is falsified and requires structural refinement.**

A model's willingness to commit to such a sharp, short-term prediction elevates its scientific status beyond a mere post-hoc fit.

6 Dark Matter and Baryogenesis

6.1 Dark Matter: Sterile Neutrino

The sterile neutrino from the seesaw mechanism, with production enhanced by the torsion phase $\lambda_T \approx 0.8095$, yields:

$$M_s \approx 0.66 \text{ keV}, \quad \theta_s^2 \approx 3.77 \times 10^{-5}, \quad \Omega_s h^2 \approx 0.12, \tag{15}$$

and decays to photons and neutrinos with lifetime $\tau_s \approx 1.90 \times 10^{17}$ years, producing an X-ray line at:

$$E_\gamma = \frac{M_s}{2} \approx 0.33 \text{ keV}. \tag{16}$$

This prediction was made before any observational constraints. Future missions such as Arcus and Lynx can test it.

6.2 Baryogenesis from SQM Washout

The radion decays to SM fermions, producing a baryon asymmetry $\eta_B^{(0)} \approx 2.61 \times 10^2$. During the SQM phase transition at $T \approx 0.2866$ eV, the SQM field washes out the asymmetry via $\Gamma_\chi = g_\chi^2 T$, giving:

$$\eta_B = \eta_B^{(0)} e^{-\Gamma_\chi / H} \approx 6.08 \times 10^{-10}, \tag{17}$$

with $g_\chi \approx 1.05 \times 10^{-5}$. The baryogenesis efficiency is thus **derived, not assumed**.

7 SQM Quantization and Dark Energy Origin

Using the heat-kernel expansion, the one-loop effective action for χ in curved spacetime with torsion yields the RG equations:

$$\mu \frac{d\Lambda}{d\mu} = \frac{m^4}{32\pi^2}, \tag{18}$$

$$\mu \frac{d}{d\mu}\left(\frac{1}{G}\right) = \frac{m^2}{96\pi^2}, \tag{19}$$

$$\mu \frac{d\xi}{d\mu} = -\frac{m^2}{96\pi^2}, \qquad \mu \frac{d\kappa}{d\mu} = -\frac{m^2}{192\pi^2}. \tag{20}$$

With $m \approx 0.0035$ eV (the seesaw scale), the cosmological constant is generated:

$$\Lambda_{\text{obs}} = \frac{m^4}{32\pi^2} \ln\left(\frac{M_{\text{Pl}}}{1 \text{ eV}}\right) \approx 3.3 \times 10^{-47} \text{ GeV}^4, \tag{21}$$

matching observations. This is a consistency condition, not a prediction—it shows the seesaw scale is naturally compatible with the dark energy scale.

8 Black Hole Entropy and Gauge Unification

8.1 Black Hole Entropy

The Wald entropy for the effective action gives the Bekenstein-Hawking area law:

$$S = \frac{A}{4G_N} \tag{22}$$

with tiny corrections:

$$\Delta S = 8\pi\alpha A R_h \approx O(10^{-60}), \tag{23}$$

consistent with black hole thermodynamics.

8.2 Gauge Unification (Open UV Question)

We have tested the Randall-Sundrum bulk with SM, MSSM, GUT multiplets, and brane-localized kinetic terms. All tested configurations yield a unification scale of $M_{\rm GUT} \approx$ 100 GeV and proton decay lifetime far below the experimental limit. The KK tower contribution is proportional to b_a, pushing the couplings apart rather than bringing them together. This is a structural issue with the RS bulk alone.

We therefore decouple the UV completion: a specific GUT model (e.g., SU(5) with SUSY) is required to achieve unification. This is left for future work. **The present work is self-contained for all IR observables.**

9 Honest Assessment of TRC V24

1. **BAO Fit:** $\chi^2/\mathrm{dof} = 1.604$. We do not claim TRC fits BAO as well as ΛCDM. The model sacrifices BAO precision to resolve S_8 and H_0, leaving a clear path for improvement (e.g., refining the Vainshtein profile or radion potential).
2. **Fitted vs Derived:** Of the $\sim 28-33$ parameters, ~ 15 are derived from the 5D action, ~ 16 are fitted to data. The derived parameters (e.g., m_χ, $\langle\chi\rangle$, g_χ) are consistency conditions—they show the framework *can* produce the right values, not that it predicts them uniquely.
3. **Gauge Unification:** Explicitly an open UV question. A dedicated calculation with a specific GUT model is required; we leave this to future work.
4. **Quantum Gravity:** SQM quantization is shown to one loop; a complete quantum gravity completion remains open. This is not a Theory of Everything.
5. **Dark Matter Prediction:** The sterile neutrino mass and X-ray line were derived before any observational confirmation, strengthening the predictive power of the framework.

Prediction	Value	Test	Timeline
θ_{13}	$11 - 12°$	DUNE, Hyper-K	2027-2028
Σm_ν	$0.02 - 0.03$ eV	CMB-S4	~ 2030
X-ray line	0.33 keV	Arcus, Lynx	$\sim$ 2030s
KK gauge boson	~ 671.5 GeV	LHC Run 3	2027-2029
KK radion	~ 1071.6 GeV	LHC Run 3	2027-2029
n_s	0.9661	CMB-S4	~ 2030
r	0.0022	CMB-S4	~ 2030
Dynamic DE	$w_a \approx +0.097$	DESI, Euclid	✓ Confirmed

Table 5: TRC V24 testable predictions.

10 Testable Predictions and Summary

The decisive test is θ_{13}. TRC commits to a sharp, short-term falsification. If DUNE finds $11 - 12°$, the framework is validated. If it finds $8.5°$, TRC is falsified. This binary outcome elevates the framework's scientific status beyond a mere post-hoc fit.

11 Conclusions

We have presented TRC V24, a geometrically-motivated IR template that resolves the biggest empirical headaches of 2026 cosmology—S_8, H_0, and dynamic DE—connects them to neutrino physics and baryogenesis via shared 5D parameters, and makes a single decisive, short-term prediction (θ_{13}). It does not claim to be a TOE; gauge unification and quantum gravity are explicitly open UV questions.

The θ_{13} prediction is the golden ticket. If DUNE confirms it, TRC will be validated in a way no phenomenological model can match. If not, it will be falsified. Either way, science progresses.

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